

Label-free quantitative (LFQ) proteomics with MaxQuant and Perseus

This tutorial is for proteomics data analysis. It consists of two parts: raw data processing with MaxQuant and statistical analysis with Perseus. Both MaxQuant and Perseus are freely available from the official website. You can also download from the link below for your convenience. You should also download the 6 raw files (n = 3 for both control and flg22 treated samples) and the Arabidopsis database for searching.

<https://drive.google.com/drive/folders/1smqXHw7dfxGp4mkEfpnuSYRehVZXqHc-?usp=sharing>

Part 1: MaxQUant Search

Step 1: load up Raw Data and assign Experiment Groups

1) Double click the MaxQuant.exe icon to open the software.

Name	Date modified	Type	Size
bin	7/26/2018 4:54 AM	File folder	
MaxQuant.exe	7/26/2018 4:53 AM	Application	196 KB

2) Click Raw Data ----> Load to

bring up the file selection window; Find and import six files. See Fig below.

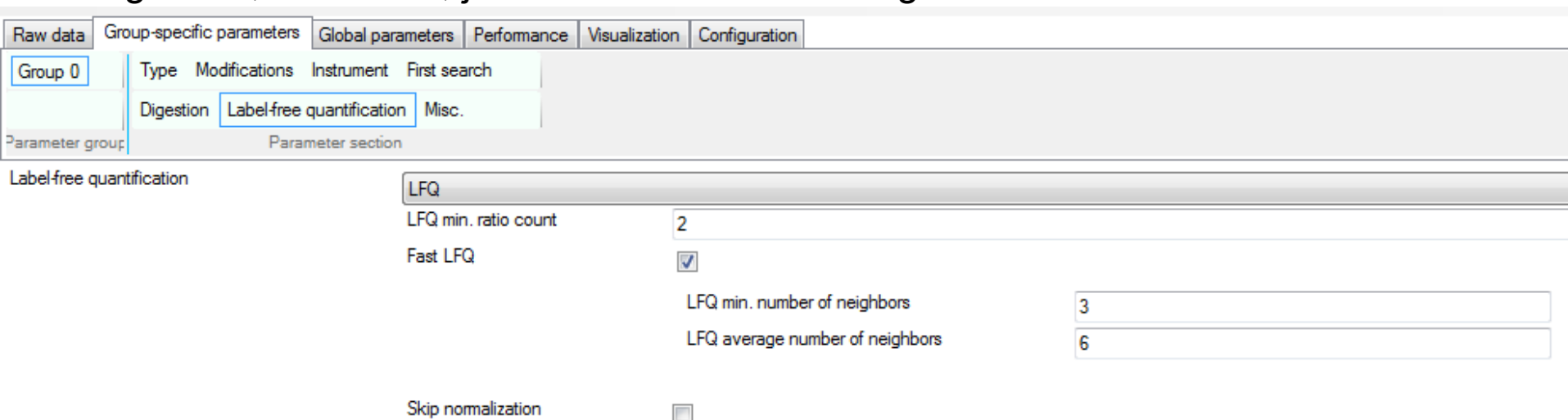
File									
	File	Exists	Size	Data format	Parameter group	Experiment	Fraction	PTM	
1	E:\MQ_Tutorial\Raw Data\Control_1.raw	True	582.7 MB	Thermo ra...	Group 0			False	
2	E:\MQ_Tutorial\Raw Data\Control_2.raw	True	551.3 MB	Thermo ra...	Group 0			False	
3	E:\MQ_Tutorial\Raw Data\Control_3.raw	True	575 MB	Thermo ra...	Group 0			False	
4	E:\MQ_Tutorial\Raw Data\flg22_1.raw	True	510.4 MB	Thermo ra...	Group 0			False	
5	E:\MQ_Tutorial\Raw Data\flg22_2.raw	True	500.8 MB	Thermo ra...	Group 0			False	
6	E:\MQ_Tutorial\Raw Data\flg22_3.raw	True	456.7 MB	Thermo ra...	Group 0			False	

3) Select the first file by single click, so the file will be highlighted in blue. Click on the “Set experiment” in the “Edit experimental design” tab. Give it an appropriate name, such as Control_1; Repeat this for all the other files. See Fig below for the results

File									
	File	Exists	Size	Data format	Parameter group	Experiment	Fraction	PTM	
1	E:\MQ_Tutorial\Raw Data\Control_1.raw	True	582.7 MB	Thermo ra...	Group 0	Control_1		False	
2	E:\MQ_Tutorial\Raw Data\Control_2.raw	True	551.3 MB	Thermo ra...	Group 0	Control_2		False	
3	E:\MQ_Tutorial\Raw Data\Control_3.raw	True	575 MB	Thermo ra...	Group 0	Control_3		False	
4	E:\MQ_Tutorial\Raw Data\flg22_1.raw	True	510.4 MB	Thermo ra...	Group 0	flg22_1		False	
5	E:\MQ_Tutorial\Raw Data\flg22_2.raw	True	500.8 MB	Thermo ra...	Group 0	flg22_2		False	
6	E:\MQ_Tutorial\Raw Data\flg22_3.raw	True	456.7 MB	Thermo ra...	Group 0	flg22_3		False	

Step 2: Set up Group-specific parameters.

- 1) Click Group-specific parameters ----> Label-free quantification
Set “Label free quantification” to “LFQ”. Keep all default parameters. See the figure below.
- 2) For all the other parameters such as Type, Modifications, Instrument, First search, Digestion, and Misc., just use the default settings.



The screenshot displays the 'Group-specific parameters' tab in a software interface. The 'Label-free quantification' section is selected, and the 'LFQ' sub-section is expanded. The parameters are as follows:

Parameter	Value
LFQ min. ratio count	2
Fast LFQ	☒
LFQ min. number of neighbors	3
LFQ average number of neighbors	6
Skip normalization	☐

Step 3a: Set up Global parameters_configuring database.

- 1) Click Global parameters ----> Sequences. This is where you set the database for searching. Within Fasta files (top pannel), click Add and load up the database file TAIR10 pep 20101214.fasta.

The screenshot shows the 'Global parameters' window with the 'Sequences' tab selected. Under the 'Fasta files' section, a table lists the loaded database file. Below the table, search parameters are configured.

	Fasta file path	Exists	Identifier rule	Description rule	Taxonomy rule	Taxonomy ID	Organism	Variation rule
1	E:\MQ_Tutorial\Database\TAIR10_pep_20101214.fasta	True	>([!s]*)	>(.*)				

Below the table, the following search parameters are set:

- Include contaminants: ☒
- Min. peptide length: 7
- Max. peptide mass [Da]: 4600
- Min. peptide length for unspecific search: 8
- Max. peptide length for unspecific search: 25
- Variation mode: None

- 2) Click on the uploaded database file to select (highlighted in blue).

Click on Identifier rule, and select “Up to first space” (see right fig).

Click on Taxonomy ID, type in 3702. This is the ID number for *Arabidopsis thaliana*.

The 'Edit identifier rule' dialog box is shown, displaying a list of parse rules. The rule '>([!s]*)' is selected, which corresponds to 'Up to first space'.

	Parse rule	Description
1	>.*([!s]*)	Uniprot identifier
2	>([!s]*)	NCBI accession
3	>IP[!s]*)	IPi accession
4	>(.*)	Everything after ">"
5	>([!s]*)	Up to first space
6	>([!s]*)	Up to first tab character

The 'Testing parse rules' window shows a table with test results for the selected identifier rule. The table has columns for Min. entry, Header, Sequence, Identifier, Description, and Taxonomy ID.

	Header	Sequence	Identifier	Description	Taxonomy ID
1	>AT1G51370.2 Symbols: F-box/RNI-lik...	MVGKKKTK...	AT1G51370.2	AT1G51370.2...	
2	>AT1G50920.1 Symbols: Nucleolar GT...	MVQYNFKRIT...	AT1G50920.1	AT1G50920.1...	
3	>AT1G36960.1 Symbols: unknown pro...	MTRLLPYKGG...	AT1G36960.1	AT1G36960.1...	

Step 3b: Set up Global parameters_other parameters. – version 1.6.1

- 1) Click “Advance identification” under the “Global parameters” tab.
Select the “Match between runs” option (see fig below)

The screenshot shows a software window with a menu bar (File, Tools, Window, Help) and a tabbed interface. The 'Global parameters' tab is active, and within it, the 'Advanced identification' sub-tab is selected. Below the tabs, there are several parameter sections. The 'Match between runs' checkbox is checked, and the 'Match time window [min]' is set to 0.7 and 'Alignment time window [min]' is set to 20. The 'Match unidentified features' checkbox is unchecked. The 'Dependent peptides' checkbox is also unchecked.

Parameter section	
Second peptides	☒
Match between runs	☒
Match time window [min]	0.7
Alignment time window [min]	20
Match unidentified features	☐
Dependent peptides	☐

- 2) For all the other parameters, just keep the default settings.

Step 3b: Set up Global parameters_other parameters. – version 1.6.5

- 1) Go to Global Parameters and select Identification
- 2) Enable the option Match between runs

The screenshot shows the MaxQuant software interface with the 'Global parameters' tab selected. Within this tab, the 'Identification' sub-tab is active. The 'Match between runs' checkbox at the bottom is checked and highlighted with a red box. Other parameters are listed in a table below.

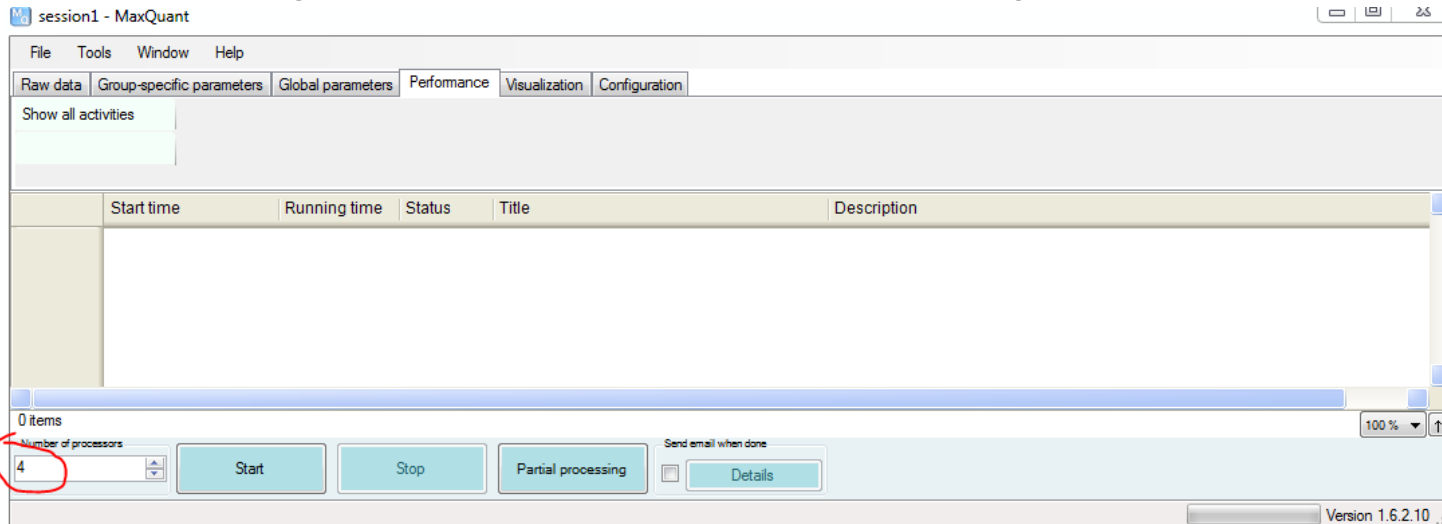
Parameter section	
PSM FDR	0.01
Protein FDR	0.01
Site decoy fraction	0.01
Min. peptides	1
Min. razor + unique peptides	1
Min. unique peptides	0
Min. score for unmodified peptides	0
Min. score for modified peptides	40
Min. delta score for unmodified peptides	0
Min. delta score for modified peptides	6
Main search max. combinations	200
Base FDR calculations on delta score	☐
Razor protein FDR	☒
Second peptides	☒
Match between runs	☒

Step 4: Set up Performance

1) Click on the “Performance” tab.

On the bottom left corner, set up the number of processors.

This depends on the processor of your computer. For instance, if you have a laptop with a quad-core processor, you can set the number to 4. Maxquant would process 4 files at the same time. The large the number, the faster the searching would be.



2) Click on “Start” and you search will begin.

3) You can monitor the progress in the “Performance” tab (fig below).

Note that 4 files are being processed here because the “number of processors” is 4.

File Tools Window Help					
Raw data Group-specific parameters Global parameters Performance Configuration					
Show all activities					
	Start time	Running time	Status	Title	Description
1	03/10/2019 11:52:23	0:53	Running	Feature_detection (1/6)	E:\MQ_Tutorial\Raw Data\Control_1.raw
2	03/10/2019 11:52:23	0:53	Running	Feature_detection (2/6)	E:\MQ_Tutorial\Raw Data\Control_2.raw
3	03/10/2019 11:52:23	0:53	Running	Feature_detection (3/6)	E:\MQ_Tutorial\Raw Data\Control_3.raw
4	03/10/2019 11:52:23	0:53	Running	Feature_detection (4/6)	E:\MQ_Tutorial\Raw Data\fig22_1.raw

TIPS

1. Reserve a lot of hard drive space for Maxquant search.

Processing proteomics raw data is very compute-intensive, and MaxQuant will write tons of output files on the disk. They are in the same folder as your raw files (see the fig below with the output data highlighted). You should have at least 30 G of free space before starting this job. MaxQuant will crash in the middle of the processing if running

Name	Date modified	Type	Size
flg22_3.index	3/10/2019 11:52 AM	INDEX File	3,935 KB
Control_2.index	3/10/2019 11:52 AM	INDEX File	4,040 KB
Control_3.index	3/10/2019 11:52 AM	INDEX File	3,914 KB
flg22_1.index	3/10/2019 11:52 AM	INDEX File	3,949 KB
flg22_2.index	3/10/2019 11:52 AM	INDEX File	3,944 KB
mqpar.xml	3/10/2019 11:52 AM	XML Document	17 KB
Control_1.index	3/10/2019 11:45 AM	INDEX File	3,903 KB
flg22_3.raw	1/12/2017 5:21 AM	Xcalibur Raw File	467,757 KB
flg22_2.raw	1/12/2017 12:23 AM	Xcalibur Raw File	512,841 KB
flg22_1.raw	1/11/2017 7:24 PM	Xcalibur Raw File	522,744 KB
Control_3.raw	1/9/2017 6:12 AM	Xcalibur Raw File	588,900 KB
Control_2.raw	1/9/2017 1:34 AM	Xcalibur Raw File	564,586 KB
Control_1.raw	1/8/2017 8:56 PM	Xcalibur Raw File	596,757 KB
combined	3/10/2019 12:02 PM	File folder	
flg22_2	3/10/2019 12:01 PM	File folder	
flg22_3	3/10/2019 12:00 PM	File folder	
Control_1	3/10/2019 11:56 AM	File folder	
Control_3	3/10/2019 11:56 AM	File folder	
Control_2	3/10/2019 11:56 AM	File folder	
flg22_1	3/10/2019 11:56 AM	File folder	

2. The time takes for the job depends on the performance of your computer. It can be anywhere between a few hour to more than 10 hours. For the quad-core processor computer tested here, it took around 2 h.

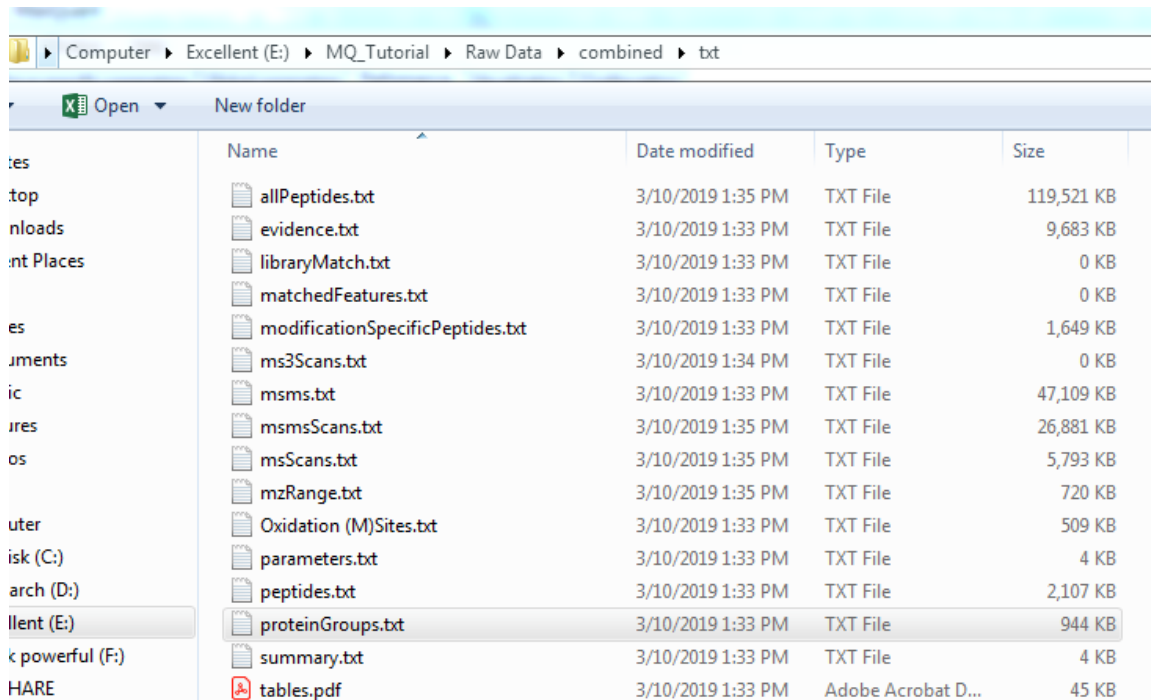
3. This tutorial shows how to perform a very basic search. Many parameters can be customized for individual projects.

Part 2: Statistical analysis with Perseus

Many software can be used for statistical analysis of proteomics. This tutorial will use Perseus for demonstration.

MaxQuant has many output files. They are located in the combined\txt folder (the same root folder as the raw files for searching, see the fig below for an example). This tutorial focuses on identifying proteins with different abundance between the control and flg22-treatment conditions. The student t test will be used for statistical test.

There are many other analysis that Perseus can perform. Abundant online resources are available for learning.

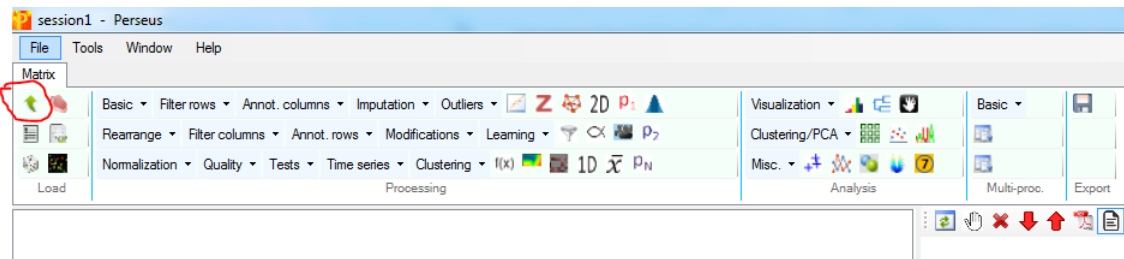


	Name	Date modified	Type	Size
tes	allPeptides.txt	3/10/2019 1:35 PM	TXT File	119,521 KB
top	evidence.txt	3/10/2019 1:33 PM	TXT File	9,683 KB
nloads	libraryMatch.txt	3/10/2019 1:33 PM	TXT File	0 KB
nt Places	matchedFeatures.txt	3/10/2019 1:33 PM	TXT File	0 KB
es	modificationSpecificPeptides.txt	3/10/2019 1:33 PM	TXT File	1,649 KB
uments	ms3Scans.txt	3/10/2019 1:34 PM	TXT File	0 KB
ic	msms.txt	3/10/2019 1:33 PM	TXT File	47,109 KB
ures	msmsScans.txt	3/10/2019 1:35 PM	TXT File	26,881 KB
os	msScans.txt	3/10/2019 1:35 PM	TXT File	5,793 KB
uter	mzRange.txt	3/10/2019 1:35 PM	TXT File	720 KB
isk (C:)	Oxidation (M)Sites.txt	3/10/2019 1:33 PM	TXT File	509 KB
arch (D:)	parameters.txt	3/10/2019 1:33 PM	TXT File	4 KB
llent (E:)	peptides.txt	3/10/2019 1:33 PM	TXT File	2,107 KB
k powerful (F:)	proteinGroups.txt	3/10/2019 1:33 PM	TXT File	944 KB
HARE	summary.txt	3/10/2019 1:33 PM	TXT File	4 KB
	tables.pdf	3/10/2019 1:33 PM	Adobe Acrobat D...	45 KB

1. Upload the ProteinGroups.txt data

a) Double click the Perseus.exe icon to open.

b) Click the “Generic matrix upload” (the green button circled in red below).



c) Click on select and upload the “proteinGroups.txt” file from MaxQuant search.

d) Select columns from "LFQ intensity Control_1" to "LFQ intensity flg22_3" to the Main box (see the fig below, left). Click OK

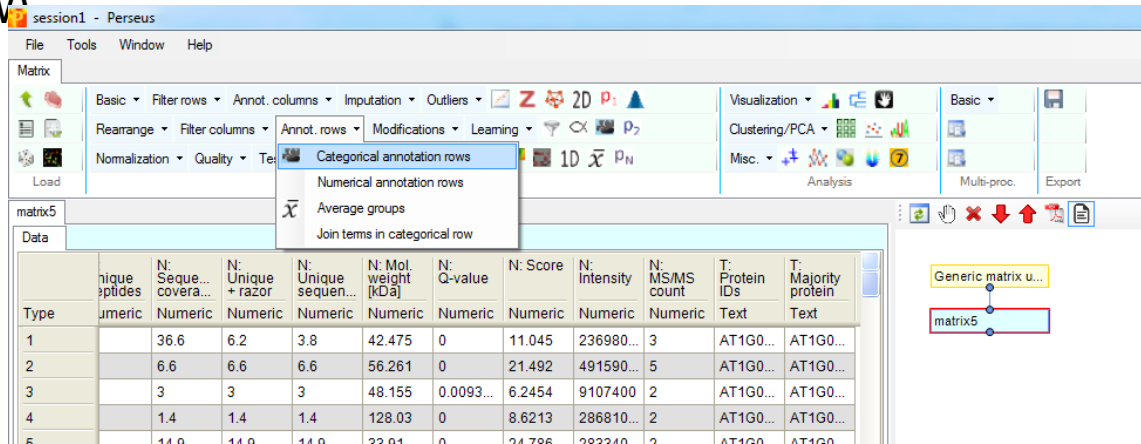
e) Perseus will generate a matrix after

The left screenshot shows the 'Select' dialog box in Perseus. The 'Main' tab is active, and a list of columns is shown on the left. The columns 'LFQ intensity Control_1', 'LFQ intensity Control_2', 'LFQ intensity flg22_1', 'LFQ intensity flg22_2', and 'LFQ intensity flg22_3' are selected. The 'OK' button is visible at the bottom right.

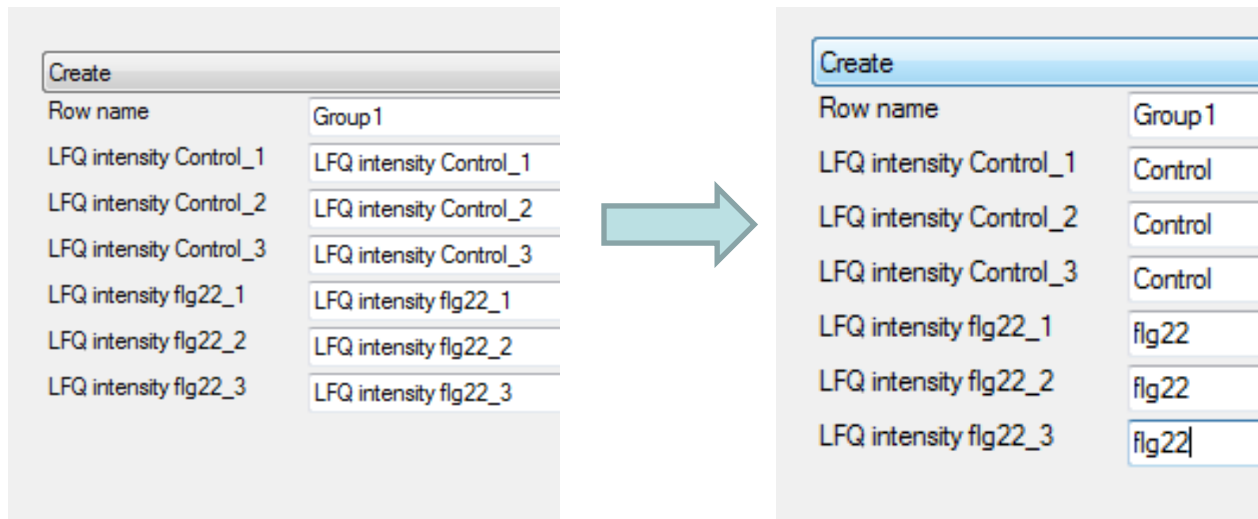
The right screenshot shows the resulting matrix data table. The table has columns for 'Type', 'Protein IDs', 'Majority protein IDs', and various numerical values. The table is titled 'matrix5' and contains 32 rows of data. The 'Protein IDs' column lists various protein identifiers, and the 'Majority protein IDs' column lists the majority protein for each row.

2. Assign experimental groups for statistical test.

a) Click on Annot.rows, select “Categorical annotation rows” from the dropdown (fig below)



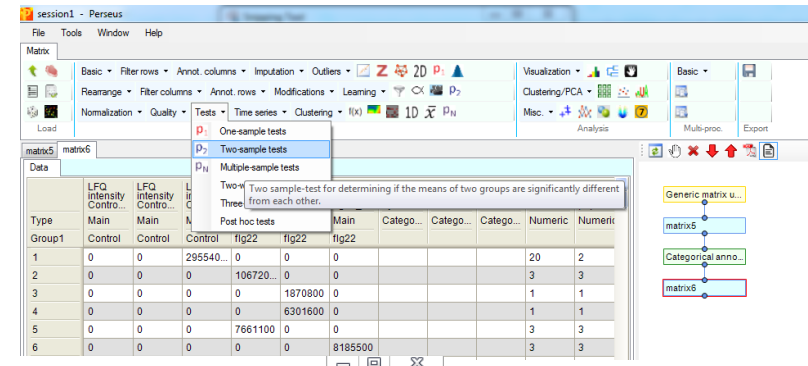
b) Change Original Grouping names (left in the following fig) to names shown in the fig below (right). Click on OK.



3. Perform statistical test

a) Click on Tests, select “two sample tests” from the dropdown menu (fig on right).

b) Set up the parameter according to the fig below. Click on OK



Two-sample tests

Cancel Suppl. table 1 OK

Grouping

Group1

First group (right)

Control

flg22

Second groups mode

Specify individual groups

Second group (left)

Control

flg22

Control

Paired

Test

Student's T-test

S0

0

Side

Both

Valid value filter

☒

Min. number of valid values

2

Min. number mode

In both groups

Min. percentage of valid values

0

Min. percentage mode

In total

Use for truncation

Permutation-based FDR

FDR

0.05

Report q-value

☒

Number of randomizations

250

Preserve grouping in randomizations

<None>

Calculate combined score

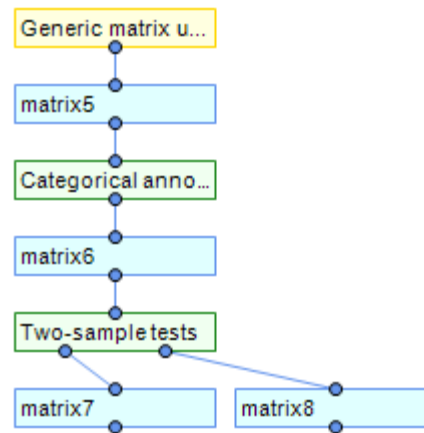
☐

Cancel Suppl. table 1 OK

4. Interpret the results

The final results is in matrix 7, which can be exported to Excel for further visualization.

Alternatively, Perseus can be used for many downstream analysis, such as PCA and clustering.



Potential questions:

1. What are protein groups? How many protein groups are identified from the dataset?
2. Acquired MS spectra are compared to theoretical spectra from the database for peptide and protein identification in shotgun proteomics. Thus, there is a chance for false discoveries. To control that, a reverse database is also searched in Maxquant. How can this information be used to calculate the false discovery rate (FDR)? What is the FDR for this dataset?
3. A volcano plot is commonly used to visualize omics data. What is a volcano plot? Generate a volcano plot for this dataset. You can use any software for this purpose.
4. If you use q-value of 0.05 to define significant proteins, how many significant proteins are there between the control and flg22-treatment samples?